PREPRINT

Calibrated Critics: Learning When to Trust Rubric Rewards in Open-Ended RL

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Code **Model** **Data** **Project**

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Abstract

Rubric-based reinforcement learning has become a standard way to post-train language models on tasks without verifiable answers, yet the judge that scores each rubric item is itself an imperfect proxy. We show that the judge's confidence is informative: rubric items on which the judge is uncertain account for most of the reward that a policy later exploits. We introduce *Calibrated Critics*, which reweights each criterion by an estimate of the judge's reliability on that criterion and prompt. Across long-form research and medical benchmarks, our method improves held-out gold judge scores by up to 6.1 points while matching the baseline's in-domain reward, and it substantially narrows the gap between training and gold judges throughout training.

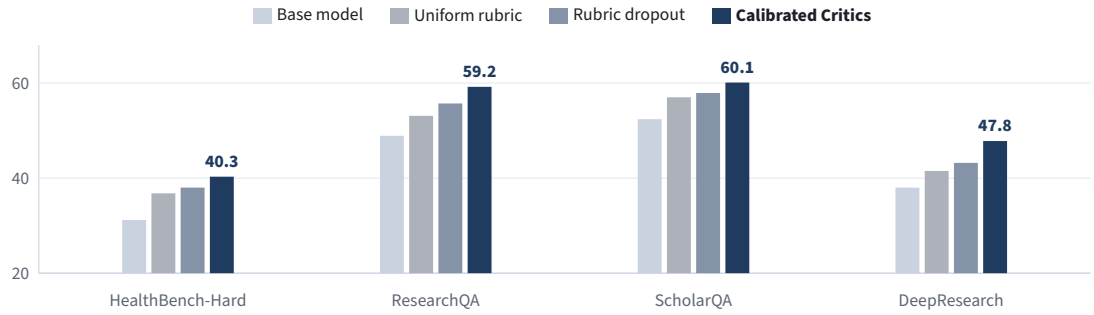


Figure 1 **Out-of-distribution results.** Gold judge scores on four long-form benchmarks. Calibrated Critics improves over every baseline on all four.

1 Introduction

Reinforcement learning with verifiable rewards works well when a ground truth exists (Shao et al., 2024). Many of the tasks we care most about, such as explaining a research area or giving medical

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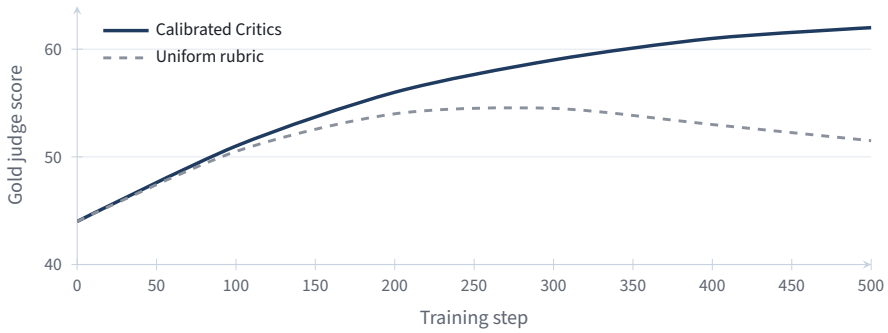


Figure 2 **Training dynamics.** Gold judge score on held-out prompts during training. The uniform rubric baseline peaks and then declines as the policy exploits unreliable criteria; reliability weighting keeps improving.

advice, are open-ended. A common recipe is to write a rubric for each prompt and let an LLM judge grade each criterion (Gunjal et al., 2025). The rubric, however, is a proxy, and a policy optimized against it long enough learns to exploit the difference (Gao et al., 2023).

Our contributions are as follows:

- We show that judge uncertainty predicts which rubric items will be exploited, before exploitation happens.
- We propose a per-criterion reliability weight that is cheap to estimate and drops into any GRPO-style objective.
- We evaluate on four long-form benchmarks and two model scales.

Main finding. Most reward hacking concentrates on a small set of rubric items where the judge was already uncertain at the start of training.

2 Method

2.1 Reliability-weighted rubric reward

Given a prompt x with rubric $\{c_k\}_{k=1}^{K}$ and weights w_k, the standard reward for a response y is the weighted fraction of satisfied criteria. We instead use

$$r(x, y) \;=\; \frac{\sum_{k=1}^{K} w_k \, \rho_k(x) \, \mathbb{I}[\, y \models c_k \,]}{\sum_{k=1}^{K} w_k \, \rho_k(x)}, \tag{1}$$

where $\rho_k(x) \in [0, 1]$ is the estimated reliability of the judge on criterion c_k.

Estimating reliability. We sample the judge m times per criterion and set ρ_k to one minus the normalized entropy of its votes.

3 Experiments

Table 1 reports the main results and Figure 2 shows the training dynamics.

Table 1 **Main results.** Gold judge scores on out-of-distribution benchmarks. Best values are shown in bold.

Method	HealthBench-Hard	ResearchQA	ScholarQA	Avg.
Base model	31.2	48.9	52.4	44.2
Uniform rubric	36.8	53.1	57.0	49.0
Rubric dropout	38.0	55.7	57.9	50.5
Calibrated Critics	**40.3**	**59.2**	**60.1**	**53.2**

References

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.

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Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

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A Implementation details

A.1 Training hyperparameters

We train with GRPO using 8 rollouts per prompt, a learning rate of 10^{-6}, and a KL coefficient of 0.001 for 500 steps.

A.2 Judge configuration

The training judge is sampled $m = 5$ times per criterion at temperature 0.7.

B Prompts

Rubric judge prompt

```
You are grading a response against a single rubric criterion.
Question: {question}
Response: {response}
Criterion: {criterion}
Answer YES if the response satisfies the criterion, otherwise NO.
```

C Additional results

Per-benchmark training curves are shown in the supplementary material.